21.11 Cancer

Cancer causes about 4 million deaths annually throughout the world. In India more than one million people suffer from cancer. Cancer is derived from Latin word meaning crab. The study of cancer is called **Oncology** (**Oncos- Tumor**).

Cancer is an abnormal and uncontrolled division of cells that invade and destroy surrounding tissue forming a tumor or **neoplasm** (new growth). It is a heterogenous group of cells that do not respond to the normal cell division.

The cancerous cells migrate to distant parts of the body and affect new tissues. This process is called **metastasis**. The frequent sites of metastasis are lungs, bones, liver, skin and brain.

More to Know

World Cancer Day - 4th February

National Cancer Awareness Day -7th November

21.11.1 Types of Cancers

Cancers are classified on the basis of the tissues from which they are formed.

1. **Carcinomas** arise from **epithelial** and **glandular tissues**. They include cancers of skin, lung, stomach and brain. About 85% of the tumours are carcinomas
2. **Sarcomas** are occur in the **connective** and **muscular tissue.** They include the cancer of bones, cartilage, tendons, adipose tissue and muscles. These form 1% of all tumours.

More to Know

Types of Tumours

Benign tumours or Non malignant tumours: Remain confined in the organ affected and do not spread to other parts of the body.

Malignant tumours: Mass of proliferating cells which grow very rapidly invading and damaging the surrounding normal tissues.

3. **Leukaemia** are characterized by an increase in the formation of white blood cells in the bone marrow and lymph nodes. Leukaemia are called **blood cancers.** Most common type of cancer which also affect children below 15 years of age.

21.11.2 Carcinogenic Agents

Cancer causing agents are called **carcinogens**. They are physical, chemical agents, ionizing radiations and biological agents.

Physical Irritant: Heavy smoking causes lung cancer and cancers of oral cavity, pharynx (throat) and larynx. Betel and tobacco chewing causes oral cancer. Excessive exposure to sunlight may cause skin cancer.

Chemical agents: Nicotine, caffeine, products of combustion of coal and oil, pesticides, asbestos, nickel, certain dyes and artificial sweetners induce cancer.

Radiations: Ionizing radiations like X-rays, gamma- rays, radioactive substances and non-ionising radiations like UV rays cause DNA damage leading to cancer.

Biological agents: Cancer causing viruses are called oncogenic viruses.

21.11.3 Treatment of Cancer

The treatment of cancer involves the following methods:

Surgery: Tumours are removed by surgery to prevent further spread of cancer cells.

Radiation therapy: Tumour cells are irradiated by lethal doses of radiation while protecting the surrounding normal cells.

Chemotherapy: It involves administration of anticancerous drugs which prevent cell division and are used to kill cancer cells.

Immunotherapy: Biological response modifiers like interferons are used to activate the immune system and help in destroying the tumors.

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